

IAM Permissions — EuroConnect Telecom

Google Cloud Storage Bucket Access Control

1. Project & Bucket Details

GCP Project ID	mlip-485615
Bucket Name	greencart-churn-regression
Bucket Location	europe-west3 (Frankfurt)
Storage Class	Standard

2. IAM Role Assignments

Team Member	Email	IAM Role
Matias Camacho (Project Lead)	matiasecam@gmail.com	Storage Admin (roles/storage.admin)
Wenyi Zhou	wy.zhou0@gmail.com	Storage Object User (roles/storage.objectUser)
Anusha Dagar	dagaranusha@gmail.com	Storage Object User (roles/storage.objectUser)
Steph Sukirno	Stephsukirno80@gmail.com	Storage Object User (roles/storage.objectUser)

All team members also have the **Service Usage Consumer** (roles/serviceusage.serviceUsageConsumer) role at the project level, which is required to make API calls to GCS from client libraries.

3. Role Explanations

Storage Admin (roles/storage.admin)

Assigned to the project lead (Matias Camacho). This role grants full control over the bucket and its objects: create/delete buckets, upload/download/delete objects, and manage IAM policies on the bucket. The project lead needs this to manage the bucket structure (inputs/, artifacts/, outputs/), grant access to teammates, and troubleshoot permission issues.

Storage Object User (roles/storage.objectUser)

Assigned to all other team members. This role allows reading and writing objects (files) within the bucket, but does not allow deleting objects, creating/deleting buckets, or modifying IAM policies. This follows the principle of least privilege: teammates can upload training data, download model artifacts, and save predictions, but cannot accidentally delete critical files or change access controls.

Service Usage Consumer (roles/serviceusage.serviceUsageConsumer)

Required at the project level for all team members. Without this role, API calls to Google Cloud Storage from the Python client library (google-cloud-storage) fail with a 403 Forbidden error, even if the user has bucket-level permissions. This role allows the user to consume GCP services (make API requests) within the project.

4. Security Rationale

The IAM configuration follows the **principle of least privilege**: each team member receives only the permissions they need to do their work.

Principle	Implementation
Least privilege	Teammates get objectUser (read/write files only), not admin (full control)
Separation of duties	Only the project lead can manage bucket structure and IAM policies
No over-provisioning	No one has project-level Owner or Editor roles; permissions are scoped to the bucket
Accidental deletion prevention	objectUser cannot delete objects, protecting training data and model artifacts